

Representations without a unit-preservation requirement

`MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation`

abbrev

Names $*$ -representations of a complex C^* -algebra on a Hilbert space when no unit equation is required.

Statement

Let A be a complex C^* -algebra considered without assuming a unit, and let H be a complex Hilbert space. `NonUnitalCStarRepresentation A H` is the type of $*$ -representations

$$\pi : A \longrightarrow \mathcal{B}(H)$$

for which no equation involving a unit is required.

Definition

For $a, b \in A$ and $z \in \mathbb{C}$, such a map satisfies

$$\pi(a + b) = \pi(a) + \pi(b), \quad \pi(za) = z\pi(a),$$

$$\pi(ab) = \pi(a)\pi(b), \quad \pi(a^*) = \pi(a)^*, \quad \pi(0) = 0.$$

Each $\pi(a)$ is a bounded complex-linear operator. Nonzero action, irreducibility, injectivity and unit preservation are separate conditions in later declarations.

Assumptions

The Hilbert space is complete. Neither A nor H is required to be separable or nonzero. A unital algebra is allowed; the abbreviation merely does not require the representation to preserve its unit.

Conclusion

The result is a type of $*$ -representations, not a theorem asserting that a chosen representation is nonzero, irreducible or faithful.

Main citations

- Exact definition and context

Lean source signature (exact)

The complete declaration below is a separate exact source excerpt; the original header record is retained with the manuscript.

```
-- A representation of a possibly genuinely nonunital complex C-star
algebra. -/
abbrev NonUnitalCStarRepresentation (A : Type u)
  [NonUnitalCStarAlgebra A] (H : Type v)
  [NormedAddCommGroup H] [InnerProductSpace ℂ H] [CompleteSpace H] :=
  A →*na [ℂ] (H →L [ℂ] H)
```

In the source	Mathematical meaning
(A : Type u) [NonUnitalCStarAlgebra A]	The complex C^* -algebra A , with no assumption requiring or forbidding a unit.
(H : Type v)	The space on which algebra elements will act.
[NormedAddCommGroup H] [InnerProductSpace ℂ H] [CompleteSpace H]	The complete complex Hilbert-space structure on H .
$H \rightarrow L[\mathbb{C}] H$	The space $\mathcal{B}(H)$ of bounded complex-linear maps from H to itself, with composition as multiplication and adjoint as star.
$A \rightarrow_{*na} [\mathbb{C}] (H \rightarrow L[\mathbb{C}] H)$	The entire right-hand side: the type of $*$ -representations $\pi : A \rightarrow \mathcal{B}(H)$, with no unit-preservation field.

Exact source and proof.

Earlier published Card and PDF

▼ Exact content identity

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